No of additional sheets 2

LOURDES MATHA COLLEGE OF SCIENCE & TECHNOLOGY

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ANSWER BOOK

Name Adwaith A M Class No. 05 Branch MCA

Subject Deep learning Semester S3

Mark Tabulation

Mark Divisions	Q.1	Q.2	Q.3	Q.4	Q.5	Q.6	Q.7	Q.8	Q.9	Q.10	Q.11	Q.12	Q.13	Q.14	Q.15
	3	2½	2		5½	3	6	½							
Total															
Grand Total													23		

2, The rectified linear unit is the most commonly used activation function in deep learning. The function returns 0 if the input is negative, but for any positive input, it returns that value back. The function is defined as:

$$\begin{cases} 0 & \text{if } x \leq 0 \\ x & \text{if } x > 0 \end{cases}$$

graph?

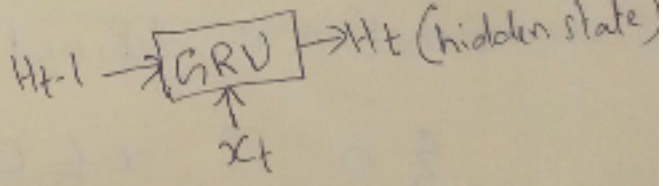
2½

It has become the default activation function for many types of neural networks because a model that uses it is easier to train and often achieves better performance.

6, GRU are a gating mechanism in recurrent neural networks. The GRU is like a long short term memory (LSTM) with a forget gate but has fewer parameters than LSTM as it lacks an output gate. GRU's performance on certain tasks was found to be similar to that of LSTM like natural language processing, speech signal modelling etc. Gated recurrent units have been shown to exhibit better performance on certain smaller and less frequent datasets. It consists of three gates and doesnot maintain an internal cell gate. The information which is stored in the internal cell state in an LSTM recurrent unit is incorporated into the hidden state of the GRU. This collective information is passed into the next GRU. The different gates of a GRU are,

3

1, Update gate.
2, Reset gate.
3, Current memory gate.



1, max pooling is a pooling operation that selects the maximum element from the region of the feature map covered by the filter. Thus the outp after max pooling layer will be feature m containing the most prominent features of

previous feature map:

6	8	2	4
5	4	3	6
3	7	9	2
4	2	1	7

max Pooling →

8	6
7	9

3

3. Vanishing gradient problem occur in RNN when a large amount of data is trained using RNN. when lot of data with error is trained the gradient error function decreases each time it is trained or as filter increases. As this process continues the error gradient will gradually tend to zero which makes it difficult to find out the errors in the next dataset. This becomes a great problem as no further error can be found out. This can be rectified using LSTM. In long short term memory there are 3 gates. The first one is forget gate it only takes the right values and avoids wrong value so the output from LSTM will always be right.

2

5. VGG stands for visual Graphic Group it is ~~called~~ said as VGG 16 because it has 16 layers. It is a continuous neural network which has 16 layers. In

each layer an input is fed and output is obtained by adding filter.

The input to first layer is 224x224 RGB image. Then it goes through filtering through layers as follow.

* 2x convolution layer of 64 channel of 3x3 kernal and same padding
* A max pooling layer of 2x2 pool size and stride 2x1
* 2x convolution layer of 128 channel of 3x3 kernal and same padding.
* 1x max pooling layer of 2x2 pool size and stride 2x2.
* 3x convolution layer of 256 channel of 3x3 kernal and same padding.
* 1x max pooling layer of 2x2 pool size and stride 2x2.
* 3x convolution layer of 512 channel of 3x3 kernal and same padding
* 1x max pooling layer of 2x2 pool size and stride 2x2
* 3x convolution layer of 512 channel of 3x3 kernal and same padding
* 1x max pooling layer of 2x2 pool size and stride 2

ADDITIONAL SHEET

Add ReLU to each layer. So negative input unit after convolution.

* 1x dense layer of 4096 units, ReLU activation
* 1x dense layer of 4096 units, ReLU activation
* 1x dense ~~softmax~~ layer of 2 units.

7, AlexNet

AlexNet is the name given to a convolution neural network. An AlexNet has 8 layers with weights, 5 convolution layers and 3 fully connected layers. It is found out that using the ReLU as activation function it accelerated the speed of the training process by almost six times. Also used the dropout layer that prevented their model from overfitting. Further the model is trained on the imagenet dataset. It has almost 14 million images across 1000 classes. It has a deep architecture so they use padding to prevent the size of feature map from reducing drastically. The input to the model in the image of size 227x227x3

Then we apply first convolution layer with 96 filter of size 11x11 with stride 4.

output = ((input - filter) / stride) + 1

op = 55 × 55 × 96

Now we apply first max pooling of size 3×3 stride 2. Then we get resulting feature of size

227 × 227 × 96

second convolution operation is applied. This time the filter size is 5×5 and have 256 such filters. The stride is 1 and padding is 2. The output size is 27 × 27 × 256

A gain we apply a max pooling of size 3×3 and stride 2. The resulting map is of shape 13×13×256.

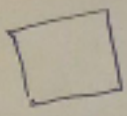
Now we apply 3rd convolution operation of filter 384 of size 3×3 stride and padding is 1. Now the output remains unchanged ie 13×13×384

After this we have final convolution layer of size 3×3 with filter 256. stride and padding is 1. Activation function is ReLU. Resulting feature map is of shape 13×13×256

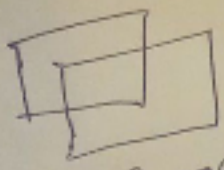
As we go deeper in Alexnet number of filter is increasing. Hence is is extracting more features as we move deeper into the architecture. Also

ADDITIONAL SHEET

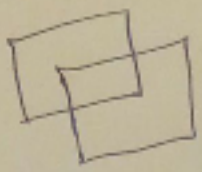
Filter size is reducing.



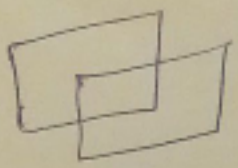
227×227×3



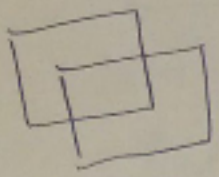
55×55×96



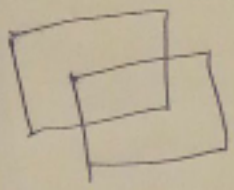
27×27×96



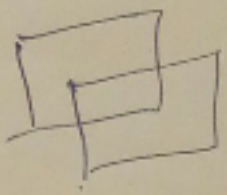
27×27×256



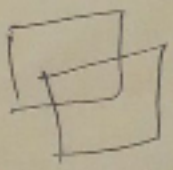
13×13×256



13×13×384



13×13×256



6×6×256

8, LSTM refers to long short term memory. There are 3 gates in LSTM.